

Task 3 Final Report

Responsible AI: Model Fairness, Bias & Explainability

1. Objective

Analyze the interpretability, fairness, and bias of a customer churn prediction model using Explainable AI techniques and recommend practical mitigation strategies.

2. Problem Statement

Machine learning models should not only be accurate but also transparent and fair. This project evaluates model behavior, explains predictions, and checks for possible bias across sensitive groups.

3. Dataset

The Telco Customer Churn dataset contains demographic information, account details, subscribed services, billing information, and churn status. It was used to train and interpret the Random Forest model.

4. Model Used

A Random Forest Classifier was selected because of its strong predictive performance. The trained model was analyzed using SHAP and feature importance techniques.

5. Explainability

Feature Importance and SHAP (SHapley Additive exPlanations) were used to explain both global model behavior and individual predictions, improving model transparency.

6. Bias Analysis

The gender attribute was selected as the sensitive feature. Churn rates for male and female customers were compared to identify potential bias. No major disparity was observed in this analysis.

7. Mitigation Recommendations

Collect balanced data, monitor fairness metrics regularly, review sensitive attributes, apply fairness-aware algorithms when required, and use SHAP/LIME to improve transparency.

8. Results

Important features included tenure, contract type, monthly charges, and total charges. SHAP explanations clearly illustrated the contribution of each feature toward individual predictions.

9. Tools & Technologies

Python, Google Colab, Pandas, NumPy, Scikit-learn, SHAP, Matplotlib, Seaborn.

10. Learning Outcomes

Learned Responsible AI concepts, model interpretability, SHAP visualization, fairness analysis, feature importance interpretation, and bias mitigation techniques.

11. Future Scope

Extend fairness analysis using additional sensitive attributes and fairness metrics, compare SHAP with LIME, and deploy explainable AI dashboards.

12. Conclusion

The project successfully demonstrated how Explainable AI and fairness analysis enhance trust, accountability, and transparency in machine learning systems.

13. Internship Requirements Checklist

Requirement	Status
Feature Importance	Completed
SHAP Explainability	Completed
Global & Local Explanations	Completed
Bias Analysis	Completed
Fairness Check	Completed
Mitigation Recommendations	Completed
Interpretation Plots	Completed
Notebook & Analysis	Completed